

RS-485 Communication Loop Troubleshooting Worksheet

Kantech EntraPass Global Edition — KT-400 / KT-1 Controller Networks

Document No.	TSW-LV-001
Title	RS-485 Communication Loop Troubleshooting Worksheet
Related Incident	Controller communication dropout — multi-site KT-400 / KT-1 deployment
Applies To	Kantech EntraPass Global Edition — KT-400 and KT-1 controllers on RS-485 daisy chain
Site / Project	_____
Date	_____
Technician(s)	_____
Revision	Rev. 1 — July 23, 2026

1. Job Snapshot

Total controllers	28 (25 × KT-400, 3 × KT-1)
Number of RS-485 loops	_____
Controllers per loop	____ / ____ / ____
Panel software / firmware version	_____
Symptom	Controllers intermittently drop offline and reconnect in EntraPass
Wiring / daisy chain topology	Confirmed — verified per SOP-LV-001/002 conventions
System history	Previously stable on prior server / software — issue began after migration to EntraPass Global Edition
Previous platform / edition	_____
New server specs (CPU / RAM / OS / virtualized Y-N)	_____
Gateway type configured	☐ Global Gateway ☐ KT-NCC ☐ Multi-Site Gateway
Migration method	☐ Clean install + reconfigured ☐ Database/config migrated from old server
Migration date	_____

2. Event Pattern Log

Record each drop event as it's reported or observed. Whether drops are simultaneous across multiple controllers or isolated to one is the single most useful diagnostic clue — it points to a shared-loop cause (ground fault, REX/motion, loop loading) versus a single-controller cause (local power supply, that controller's connections).

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3. Diagnostic Checklist

System was stable on the prior server/software and issues began only after migrating to EntraPass Global Edition. This points to the software/gateway layer, not the field wiring — work through Section G first. Sections A–F are retained for reference and to formally rule out hardware, but are lower priority given the system's prior stability.

G — EntraPass Global Edition / Server-Side (start here — system was stable pre-migration)

- ☐ Confirm which gateway architecture is configured: Global Gateway, KT-NCC, or Multi-Site Gateway — verify it matches the intended topology, not a default left over from setup.
- ☐ Compare Communication Timing settings (Devices → Connection Configuration → Setting up Communication Timing) against the values used on the prior server — polling interval/timeout/retry defaults differ by edition and can be too aggressive for a 28-controller deployment.
- ☐ Check new server resource utilization (CPU, RAM, disk I/O) during drop events — Global Edition's SQL Server-backed architecture is heavier than Special/Corporate Edition and can starve polling under load.
- ☐ Confirm antivirus/firewall/Windows Defender on the new server is not scanning or throttling the Gateway service's serial/IP traffic — add exclusions for EntraPass processes and ports if needed.
- ☐ Check EntraPass event log / Gateway statistics for repeated firmware reload or 'controller update' events — a failed or looping firmware push during migration can cause cyclic dropouts.
- ☐ Verify controller/door licensing covers all 28 controllers under the new Global license — under-licensing can cause controllers to be intermittently dropped.
- ☐ If configuration was migrated (not a clean install), check for duplicate controller addresses, leftover COM port assignments, or mismatched loop/gateway assignments carried over from the old system.
- ☐ If communication changed from RS-232/direct to Ethernet polling as part of the move, verify network path (switch ports, VLAN, latency) between the server and the field controllers.
- ☐ Check Windows Event Viewer on the new server for Gateway service crashes/restarts that would correlate with controller drop timestamps.

A — Lock Power / Ground Fault (lower priority — wiring already verified stable pre-migration)

- ☐ Measure lock power circuit for each door for shorts to ground (KT-400 supplies wet lock power directly off the controller — a ground fault here can drag down the entire 485 loop, not just that door).
- ☐ Disconnect lock power at each door sequentially and observe whether the loop stabilizes — isolates which door circuit is the source.
- ☐ Inspect strike/maglock wiring at door frames for pinched, chafed, or moisture-exposed conductors.
- ☐ Verify lock power is fused/sized correctly per KT-400 installation guide and not shared across an excessive number of doors.

B — REX / Motion Detector Correlation

- ☐ Cross-reference EntraPass event log timestamps against REX/motion activations on all doors in the affected loop.
- ☐ If drops correlate with REX/motion triggers, temporarily disable the suspect REX input and monitor for 24–48 hrs.
- ☐ Check REX/motion wiring for shared ground/shield with the 485 line — reroute if run in the same conduit or bundle.

C — Loop Loading / Topology

- ☐ Confirm number of controllers per individual 485 loop off the gateway/KT-NCC — heavy loops (20+ panels) are prone to this failure mode; split into additional loops if needed.
- ☐ Confirm true daisy-chain (no stars, no T-taps, no stub runs off the main line).
- ☐ Confirm cable length per loop is within Kantech's published maximum for 485 runs.

D — Termination & Bias

- ☐ Confirm 120Ω termination resistor is installed only at the two physical ends of each loop.
- ☐ Confirm no mid-chain controller has termination enabled (jumper/DIP switch).
- ☐ Confirm bias resistors are not duplicated across multiple controllers on the same loop.

E — Grounding

- ☐ Confirm 485 shield is grounded at one point only (typically the gateway/panel end) — not at each controller.
- ☐ For loops spanning multiple buildings or electrical services, check for ground potential difference between panels.
- ☐ Confirm controller chassis grounds are bonded per code and not floating.

F — Power Supply

- ☐ Measure DC voltage at each controller under load (door strikes firing), not just at idle — sag under load indicates undersized/failing supply.
- ☐ Confirm supply is within Kantech's specified 12–13.8 VDC range at the controller terminal, not just at the source.
- ☐ Check for shared/daisy-chained power supplies feeding multiple controllers beyond rated capacity.

4. Per-Loop Electrical Log

Complete for each RS-485 loop. Compare voltage/resistance readings across loops to spot the outlier.

Loop #	# Controllers	Termination OK (Y/N)	End-to-end resistance (Ω)	Voltage at last panel (VDC)	Notes

5. Root Cause & Resolution

Root cause identified	
Corrective action taken	
Monitoring period post-fix	
Result confirmed stable (Y/N + date)	

6. Sign-Off

Technician Signature	Date
Reviewed By	Date